



BMI estimation based on weight, height and gender.

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Introduction

Body Mass Index (BMI) is a widely recognized measure used to assess whether an individual's weight is appropriate for their height.

It is commonly employed in healthcare to categorize individuals into groups such as underweight, normal weight, overweight, and obese.

This project aims to demonstrate how machine learning can be applied to estimate BMI values, thereby reinforcing the practical use of regression models in real-world scenarios.

Problem statement:

The primary objective of this project is to estimate BMI using machine learning methods.

The model is designed to predict BMI values based on three input features: gender, height, and weight.

In addition to numerical prediction, the project also classifies BMI into health categories, making the output more interpretable and useful for end users.

Dataset and preprocessing

- The dataset used in this project consists of three features: gender, height, and weight. Gender is encoded numerically, with 1 representing male and 0 representing female. Height is measured in centimeters, while weight is measured in kilograms. The target variable is BMI, which is calculated using the standard formula. This dataset provides the foundation for training and evaluating the regression model.
- Preprocessing involved calculating BMI from the given height and weight values using the formula : $BMI = (weight) / [(height/100)^2]$.
- Gender was encoded numerically to ensure compatibility with machine learning algorithms.
The dataset was then split into training and testing subsets, with 80% allocated for training and 20% reserved for testing. This ensured that the model's performance could be evaluated on unseen data.

Methodology and model selection

The project employed a supervised machine learning approach using linear regression. The dataset was preprocessed, split into training and testing subsets, and evaluated with performance metrics such as Mean Squared Error and R^2 score to validate accuracy.

Linear Regression was chosen as the primary model for this project; This decision was based on the simplicity, interpretability, and efficiency of the method.

Since BMI is mathematically defined by a linear relationship between weight and height squared, Linear Regression is well-suited to capture this relationship.

While other models such as K-Nearest Neighbors, Decision Trees, and Random Forests were considered, Linear Regression provided the most direct and transparent approach.

Training the model

The model was trained using scikit-learn's LinearRegression implementation.

During training, the algorithm learned coefficients for each feature—gender, height, and weight—along with an intercept term. These coefficients represent the contribution of each feature to the predicted BMI.

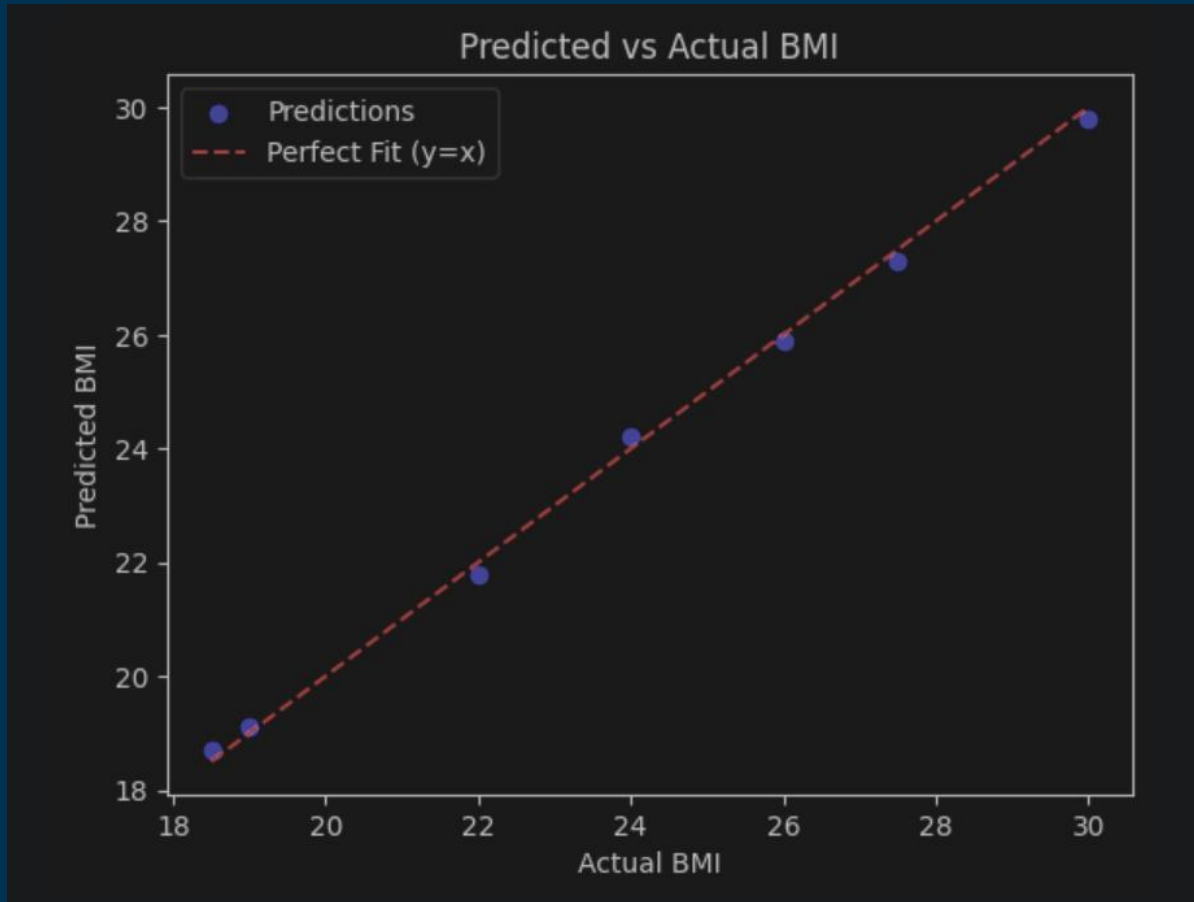
The resulting regression equation provides a clear mathematical representation of the model's understanding of the data.

The regression equation is :

$$\text{BMI} = (m1 \cdot \text{Gender}) + (m2 \cdot \text{Height}) + (m3 \cdot \text{Weight}) + c$$

- $m1$, $m2$, $m3$ are the coefficients learned during training
- C is the intercept term.

Evaluation



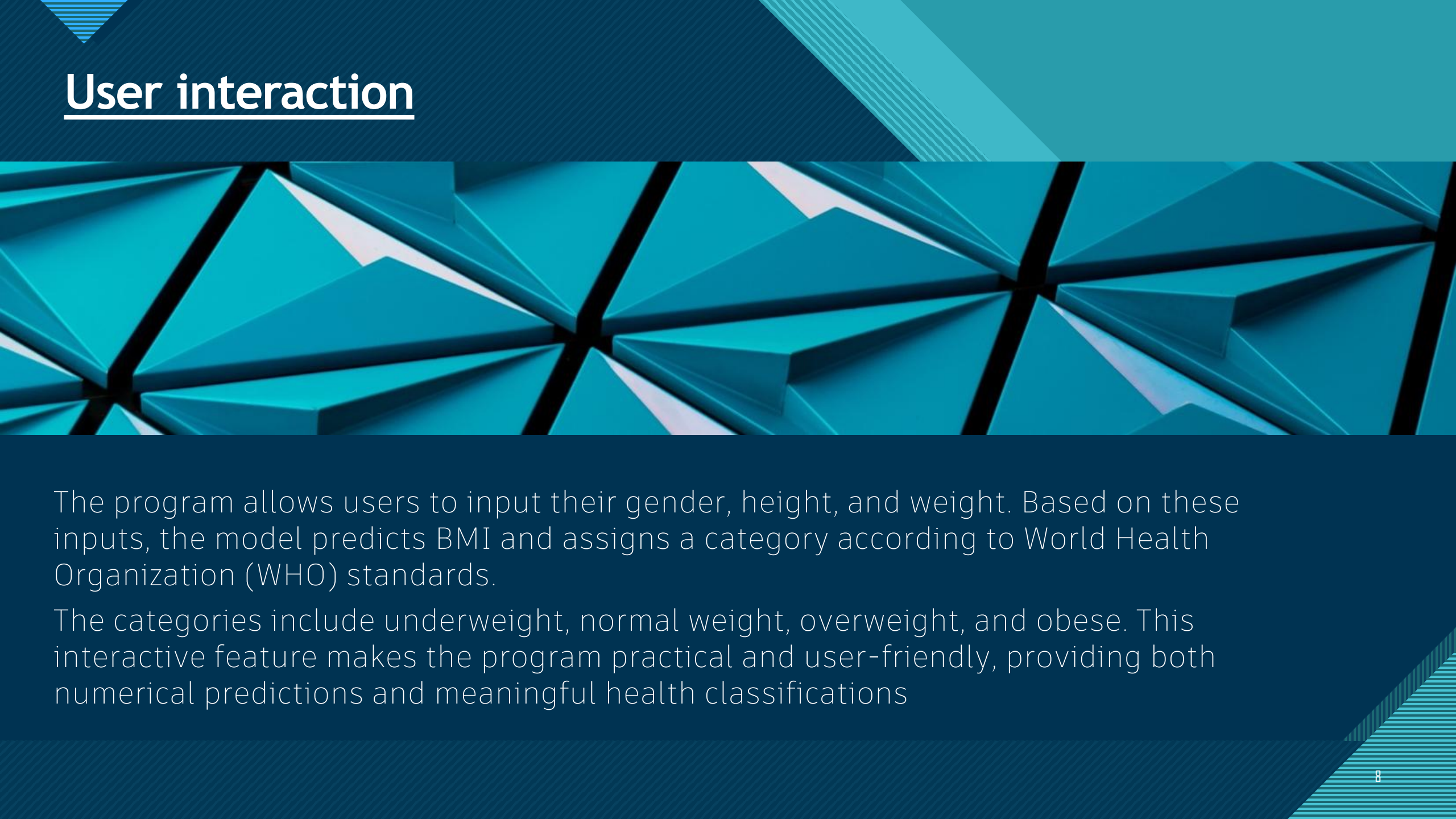
The model was evaluated using two key metrics: Mean Squared Error (MSE) and R^2 score.

MSE measures the average squared difference between predicted and actual BMI values, while R^2 indicates how well the model explains variance in the data.

The evaluation results showed a low MSE and an R^2 score close to 1.0, confirming that the model captured the relationship effectively.

Predicted versus actual BMI values were closely aligned, demonstrating high accuracy.

User interaction



The program allows users to input their gender, height, and weight. Based on these inputs, the model predicts BMI and assigns a category according to World Health Organization (WHO) standards.

The categories include underweight, normal weight, overweight, and obese. This interactive feature makes the program practical and user-friendly, providing both numerical predictions and meaningful health classifications

Conclusion and key findings

The project demonstrated that Linear Regression is highly effective in estimating Body Mass Index (BMI) from gender, height, and weight.

The model achieved a low Mean Squared Error (MSE) and an R^2 score close to 1.0, confirming that it captured the relationship between the input features and BMI with high accuracy. Predictions closely matched actual BMI values, validating the reliability of the approach.

The user interaction component successfully classified BMI into categories defined by the World Health Organization, making the output practical and meaningful.

A key insight is that while Linear Regression provided excellent results due to the formula-driven nature of BMI, the metric itself has limitations as a health indicator, since it does not account for muscle mass, fat distribution, or age. This finding emphasizes the importance of combining BMI with other health measures for a more comprehensive assessment.

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Thank You.